

K-Means Clustering Tutorial

Subject: Machine Learning Topic: Unsupervised Learning

Overview

Algorithm steps, initialization methods, elbow method, silhouette score

Objective

- K-means partitions numeric observations into k clusters by minimizing the sum of squared distances to assigned centroids.
- Each cluster centroid is the coordinate-wise mean of its assigned points.
- Because distance depends on scale, standardize features when their units or ranges differ.

Algorithm

- Choose k initial centroids, often with `k-means++` to spread starting points.
- Assign every observation to its nearest centroid, then recompute each centroid as its cluster mean.
- Repeat assignment and update until assignments stop changing, centroid movement is small, or an iteration limit is reached.

Choosing k and evaluating

- The elbow method plots within-cluster sum of squares against k and looks for diminishing improvement.
- Silhouette score compares within-cluster cohesion with separation from the nearest other cluster; values near 1 indicate clearer assignments.
- Run multiple initializations because the objective can converge to different local minima.

Limitations

- K-means is sensitive to outliers and favors roughly spherical, similarly sized clusters under Euclidean distance.
- Empty clusters need a defined reinitialization policy.
- Use domain judgment and visual inspection; a metric alone does not establish that clusters are meaningful.